

# Child pronunciation attempts — BE reference

Mobile app logs every letter/word pronunciation verification to Supabase for analytics and QA.

**Table:** `public.child_pronunciation_attempt`

| Column                          | Type       | Description                                                                    |
|---------------------------------|------------|--------------------------------------------------------------------------------|
| <code>id</code>                 | uuid       | PK                                                                             |
| <code>child_id</code>           | uuid       | FK → <code>child_user.id</code> (cascade delete)                               |
| <code>session_id</code>         | text       | Groups retries in one lesson step, e.g. <code>pron-1782121407706-vvlecr</code> |
| <code>attempt_number</code>     | int        | 1- based retry index within the session                                        |
| <code>attempt_type</code>       | text       | <code>letter</code> or <code>word</code>                                       |
| <code>target_display</code>     | text       | UI label shown to child (Arabic letter char or word text). Nullable            |
| <code>target_sent_to_api</code> | text       | Value sent to verification API (mapped letter code or word string)             |
| <code>is_correct</code>         | boolean    | Parsed from model <code>result</code> . Null on API/network error              |
| <code>model_output</code>       | jsonb      | Raw verification API response (or <code>{ error: "..."</code> } on failure)    |
| <code>client_context</code>     | jsonb      | Request/debug metadata from the app (see below)                                |
| <code>audio_storage_path</code> | text       | Path inside storage bucket. Null if upload failed                              |
| <code>recorded_at</code>        | timestampz | When the attempt was recorded                                                  |
| <code>created_at</code>         | timestampz | Row insert time                                                                |

**Indexes:** `child_id` , `session_id` , `recorded_at` desc

**RLS:** Enabled. Authenticated parents can SELECT/INSERT only for their own children (`child_user.parent_id = auth.uid()` ).

**Migration:** `supabase/migrations/20260622_child_pronunciation_attempt.sql`

## Audio files: bucket `pronunciation-recordings`

- **Private bucket** (not public).
- **Path format:** `{child_id}/{session_id}/{attempt_number}.wav`
- Example: `7344cb1e-0d6d-4f52-af0a-a2de6f546b53/pron-1782121407706-vvlecr/1.wav`
- **Format:** WAV, 44.1 kHz, mono.
- **Storage RLS:** Any authenticated user can read/write this bucket (per-bucket open policy).

To play a file, generate a signed URL (do not store full URLs in DB):

```
-- via Supabase client
supabase.storage.from('pronunciation-recordings').createSignedUrl(path, 3600)
```

## When rows are created

Triggered after each pronunciation check on a **real device** (simulator skips save).

**Flow:**

1. Child records audio → sent to external verification API.
2. WAV uploaded to `pronunciation-recordings` (if local file exists).
3. Row inserted into `child_pronunciation_attempt`.

**App code:** `src/services/pronunciation.service.ts` · `src/hooks/useVoiceVerifier.tsx`

**Screens:** Letter pronunciation, Word pronunciation flows.

## `model_output` (jsonb)

Raw JSON from the verification APIs:

| API    | Endpoint                                                                      | Notable fields                                                                                                                                    |
|--------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Letter | <code>https://arabic-letters-api-d26k2plh4q-ew.a.run.app/verify_letter</code> | <code>result</code> , <code>predicted_letter</code> , <code>predicted_probability</code> , <code>target_probability</code> , <code>message</code> |
| Word   | <code>https://arabic-words-api-d26k2plh4q-</code>                             | <code>result</code> , <code>similarity</code> , <code>message</code>                                                                              |

ew.a.run.app/verify\_word

On failure: { "error": "<message>" } .

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## client\_context (jsonb)

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App-side metadata (not from the model):

```
{
  "endpoint": "https://...",
  "httpStatus": 200,
  "apiLatencyMs": 1234,
  "targetTextApp": "i",
  "requestPayload": { "target_letter": "alif", "threshold": "0.6" },
  "recordingDurationMs": 4500,
  "speechDetected": true,
  "peakMeteringDb": -18,
  "stopReason": "timer_expired",
  "platform": "ios",
  "appVersion": "3.2.60",
  "deviceModel": "iPhone 15",
  "isRealDevice": true,
  "isSimulator": false
}
```

Word attempts also include `fuzzy_match` / `fuzzy_threshold` in `requestPayload` .

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## Useful queries

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```
-- All attempts for a child, newest first
select * from child_pronunciation_attempt
where child_id = '<uuid>'
order by recorded_at desc;

-- Retries in one lesson step
select * from child_pronunciation_attempt
where session_id = 'pron-1782121407706-vvlecr'
order by attempt_number;

-- Attempts missing audio
select id, session_id, attempt_number, audio_storage_path
from child_pronunciation_attempt
where audio_storage_path is null;
```

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## Notes for BE

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- `session_id` is client-generated, not a DB session.
- One row per verification call, not per tap (retries = new `attempt_number` , same `session_id` ).
- `audio_storage_path` is a storage key, not a URL — resolve at read time.
- Table insert can succeed even when `audio_storage_path` is null (upload failed separately)